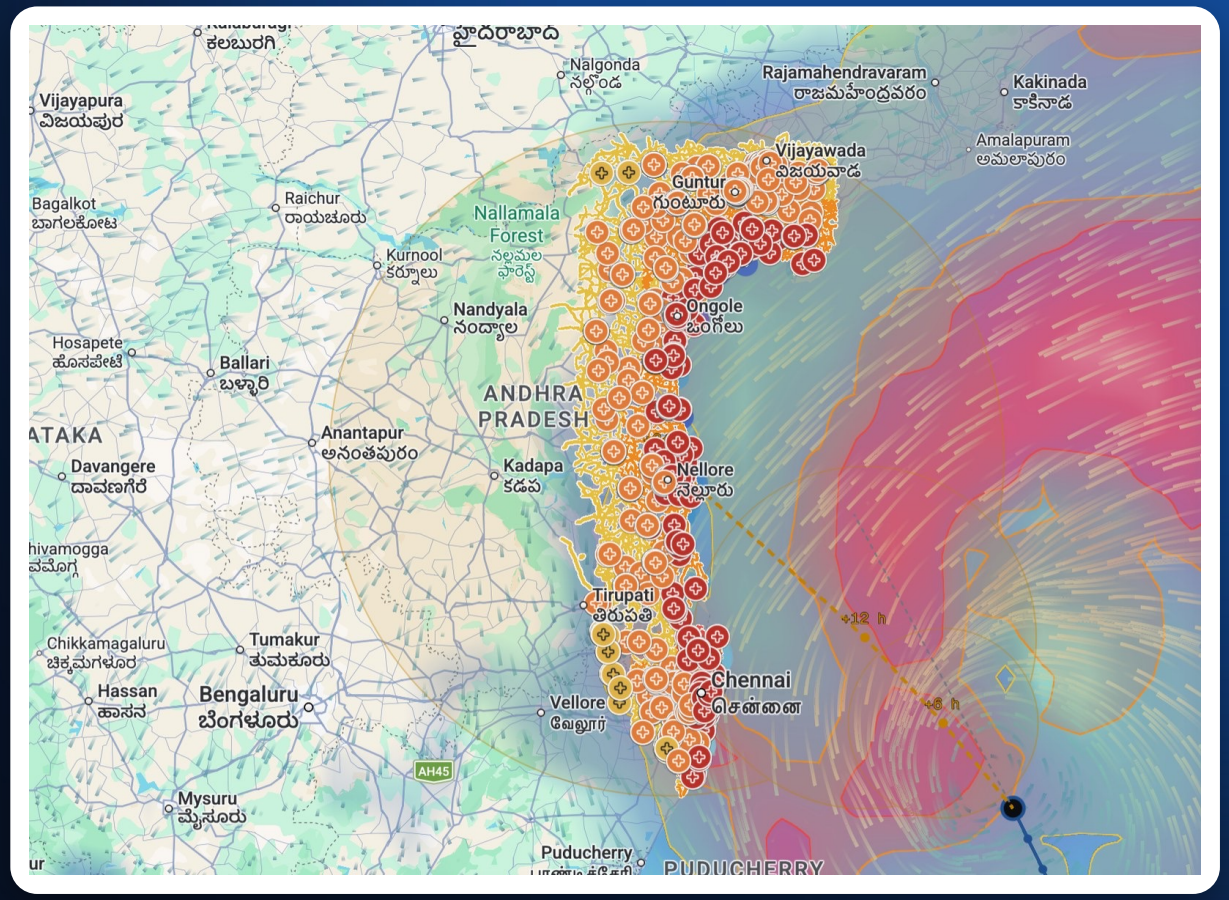
BUILD WITH AI: CODE FOR COMMUNITIES 2.0

SurgeSight

See a cyclone’s impact before it reaches the coast.

Problem Statement 05 · Track-Based Cyclone Impact & Infrastructure Vulnerability Forecaster



Built with **Gemini** **Earth Engine** **Google Maps** **Cloud Run**

TEAM ECHO

Charan · Vishal

Forecasts tell us where a cyclone will go. They don’t tell an officer what it will hit.

THE PROBLEM

IMD issues good cyclone forecasts. But before landfall, a District Collector and every department still work out by phone calls and spreadsheets what that track means for their schools, hospitals, power lines, roads and people.



The missing layer is consequence.

THE CHALLENGE

Turn one cyclone forecast into answers every department can act on, before landfall:

- Which schools and hospitals will be hit hardest, and why?
- Which roads to a hospital could be cut off?
- What changes if the track shifts 30 km north?
- What should Health do, and what should Electricity do?
- How do we warn people in Tamil, Telugu or Odia?

OUR SOLUTION

To solve it, we built SurgeSight: from one cyclone bulletin to approved action, in five steps.



1 Read

Gemini reads the IMD bulletin: text, PDF or photo

Gemini

2 Score

Every facility and main road is scored for wind, surge and rain

Earth Engine

3 Show

Alerted schools, hospitals and roads on the map, each with its reason

Google Maps

4 Advise

Gemini writes each department's advisory in its own language

Gemini

5 Approve

An officer approves, then alerts go out by email and phone

Gmail · Firebase

Numbers come from our models. Words come from Gemini. Every number Gemini writes is checked against our data.

CASE STUDY

To study and test SurgeSight, we replayed Cyclone Michaung (December 2023) bulletin by bulletin.

Severe Cyclonic Storm Michaung crossed the Andhra Pradesh coast near Bapatla on 5 December 2023, after extreme rain flooded Chennai.

WHY WE CHOSE IT

Two states, two languages

Tamil Nadu and Andhra Pradesh, dense cities and long coast

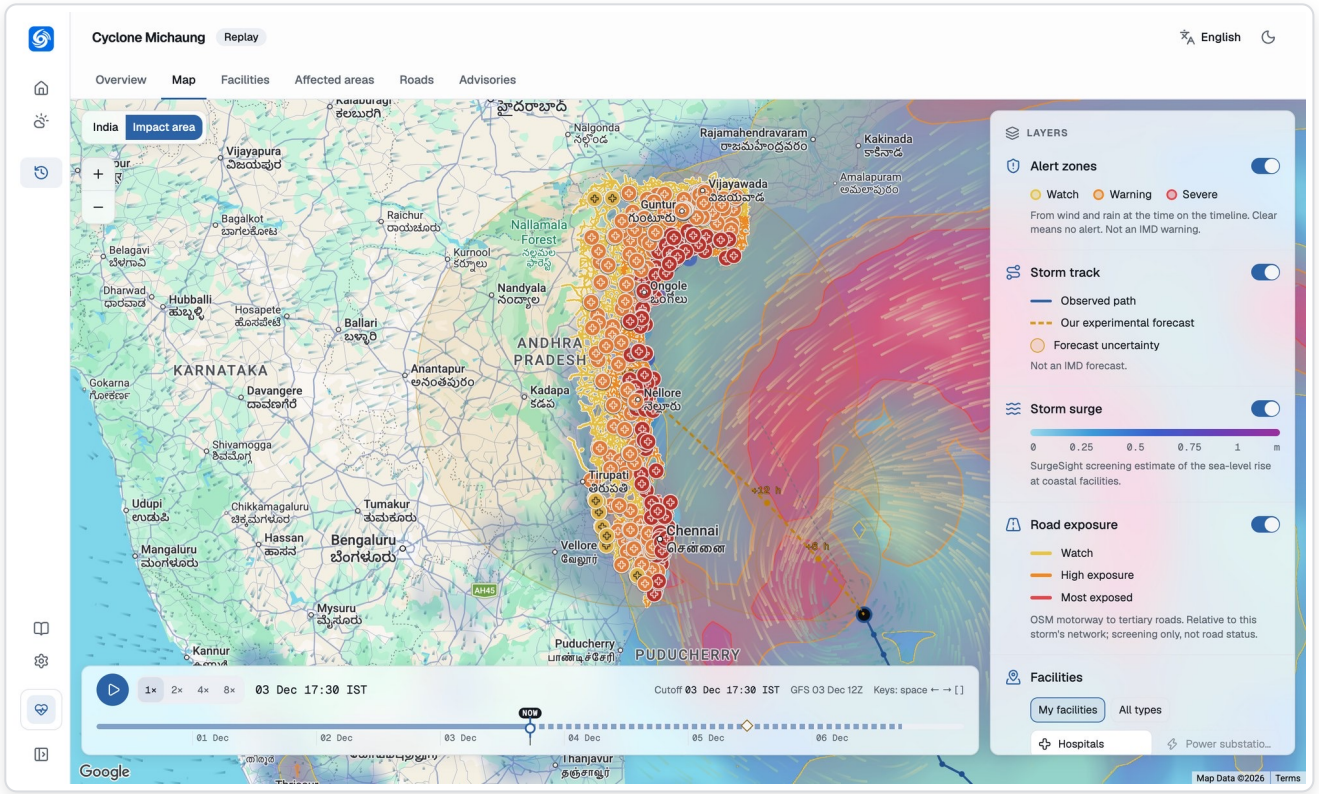
Open data to check ourselves

IBTrACS track, NOAA GFS, NASA IMERG, Sentinel-2 floods

An honest replay

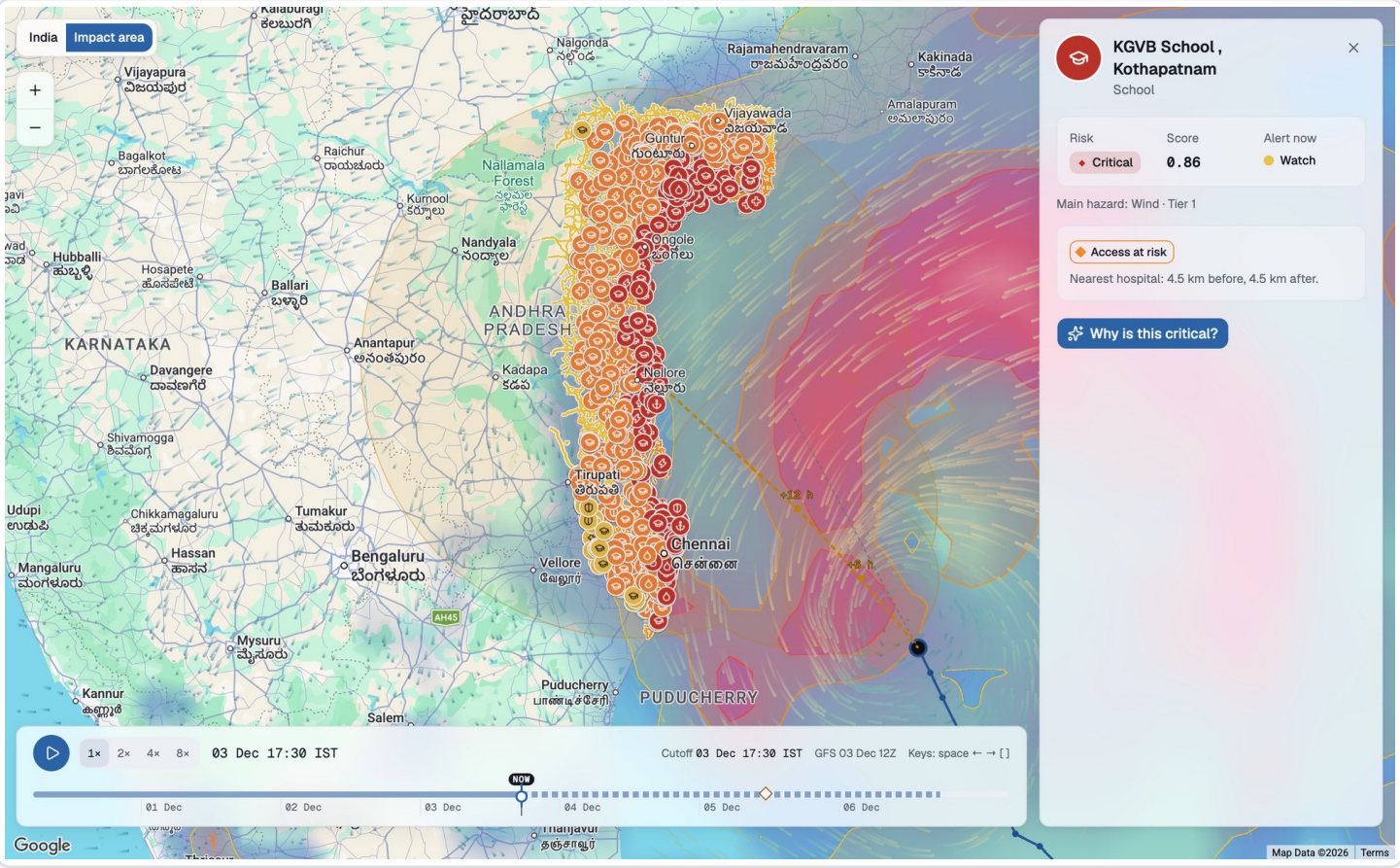
Each screen shows only what was known at that bulletin

4,214	critical facilities	**33,930 km**	main roads
10.5 M	people in priority area	**7**	departments



Our replay of Michaung on Google Maps at the 03 Dec 17:30 IST bulletin.

Every school, hospital and substation under the storm, on one map, coloured by risk.



Tap any facility (here KGVB School, Kothapatnam) to see why it was alerted.

4,214 facilities, 8 types

Schools, hospitals, substations, shelters, police, fire, water, harbours, from Low to Critical

Alert zones

Watch, warning and severe areas from wind and rain, moving with the storm

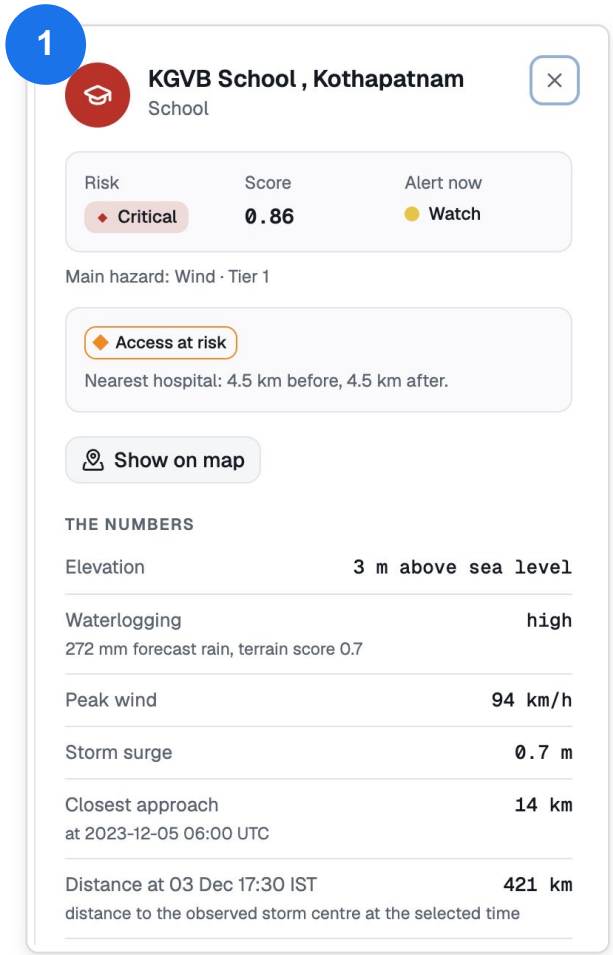
Track and forecast

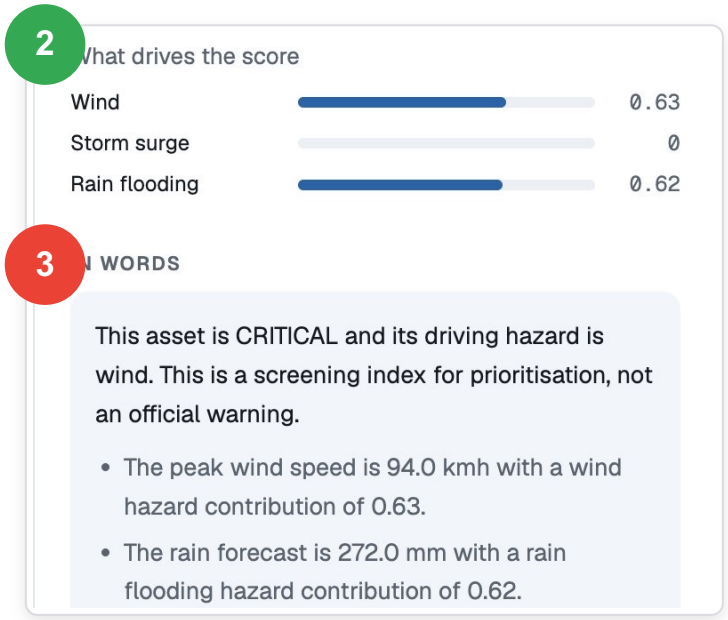
Observed path, our experimental forecast and its uncertainty

Roads to check

33,930 km of main roads by exposure; hospitals whose access is cut off

Every alert comes with its reason: the numbers first, then Gemini's words.





Road access too

Access at risk: the nearest hospital is 4.5 km away, and the main roads to it are exposed.

1 The numbers

Our models and Earth Engine: 3 m above sea level, 272 mm of forecast rain, 94 km/h peak wind, 14 km from the track.

2 What drives the score

Wind 0.63 and rain flooding 0.62 make this school Critical, in the top priority tier.

3 In words, by Gemini

A plain-language reason for the officer. Every number Gemini writes is checked against the table; any it cannot back up is removed.

FEATURE 3 · GEMINI AT WORK

What Gemini does in SurgeSight.

Reads

Turns an IMD bulletin (text, PDF or photo) into a storm track.

Explains

Tells officers why each school or hospital was alerted, and what changes in a what-if.

Advises

Writes an action plan for each of 7 departments, from the ranked facilities and roads.

Speaks six languages

English, Hindi, Tamil, Telugu, Bengali and Odia, with audio from Cloud Text-to-Speech.

Reasons with tools

An agent team built with ADK picks from 29 tools to answer questions like “what if?”

Searches

Finds the latest bulletins and news with Google Search, with sources.

Gemini 3.7 Flash on Vertex AI · It never calculates the risk: our models do. A guard removes any number Gemini cannot back up.

Each department gets its own advisory, in its own language, sent only when a person approves.



CRITICAL Health Department

புயல் எச்சரிக்கை: சுகாதாரத்துறை மருத்துவமனைகளுக்கான அவசர இடர் தணிப்பு வழிகாட்டுதல்

மதிப்பிடப்பட்ட 2200 மருத்துவமனை அமைப்புகளில் 1070 அமைப்புகள் CRITICAL இடர் வகுப்பிலும் 1070 அமைப்புகள் HIGH இடர் வகுப்பிலும் அடையாளம் காணப்பட்டுள்ளன. மதிப்பீட்டின்படி 108 சுகாதார வசதிகள் துண்டிக்கப்பட்ட நிலையிலும் 1331 வசதிகளுக்கான அணுகல் இடரிலும் உள்ளன. சென்னை மற்றும் சுற்றியுள்ள முக்கிய சாலைகளில் 1696.7 கி.மீ மிக அதிக வெளிப்பாட்டு அபாயத்தில் (most exposed) உள்ளன.

அதிக இடர் எதிர்கொள்ளும் மருத்துவமனைகளில் மின்சார ஜெனரேட்டர் எரிபொருள், டயாலிசிஸ் மற்றும் தீவிர சிகிச்சைப் பிரிவு (ICU) தொடர்ச்சியை உடனடியாக உறுதி செய்யவும்.

முன்னுரிமைச் செயல்கள்

1. Punarjeevan Hospital, Dhanvantri Multi Speciality மற்றும் Dr.Kamakshi Memorial Hospital ஆகியவற்றில் ஜெனரேட்டர் எரிபொருள் இருப்பை உறுதிசெய்க.
2. துண்டிக்கப்பட்டதாகக் குறிப்பிடப்பட்டுள்ள Pulicat Government Hospital, RMS Ropheka Multi Speciality Hospitals மற்றும் Vasantha Hospital ஆகியவற்றில் அவசர நோயாளிகள் இடமாற்ற திட்டங்களைச் செயல்படுத்துக.
3. அணுகல் இடர் உள்ள Government Hospital, Area Hospital மற்றும் Sri Mundarapottamma

Health Department advisory, written by Gemini in Tamil.

Gemini drafts
7 departments × 6 languages

Audio + CAP 1.2 alert
Cloud Text-to-Speech · standard alert format

An officer approves
Nothing is sent automatically

Sent to the team
Gmail and phone push (Firebase)

HOW IT WORKS

Our models compute the risk. Gemini explains it. Both are checked against real data.

DATA IN

IMD bulletin

Earth Engine · 10 datasets

OpenStreetMap · 4,214 facilities

→

RISK MODELS

Wind · surge · rain flooding

Road exposure · hospital access

100-run ensemble · what-if

→

GEMINI AGENTS

Gemini 3.7 Flash + ADK

29 tools

Number guard

→

OUT

Map with reasons

Advisories · 6 languages

Approved alerts

CHECKED AGAINST WHAT REALLY HAPPENED IN MICHAUNG

27.6 km

average error of our 6-hour track forecast

0.71

match between forecast rain and NASA's observed rain

2×

satellites saw water twice as often on the roads we flagged

BUILT ON GOOGLE

23 Google services, running on Cloud Run today.

AI

- Gemini 3.7 Flash
- Vertex AI
- Gemini Live
- Agent Development Kit
- Search grounding
- Text-to-Speech
- Speech-to-Text

MAPS & EARTH

- Earth Engine
- Maps JavaScript
- Places
- Routes
- Geocoding
- Weather

DATA

- BigQuery GIS
- Firestore
- Cloud Storage

ALERTS

- Gmail API
- Firebase Messaging
- Firebase Auth

PLATFORM

- Cloud Run
- Cloud Scheduler
- Secret Manager
- Cloud Build
- Artifact Registry

Live on Cloud Run

- **One container, asia-south1**
  Scales to zero; scales out on a storm day
- **Weather refreshes itself**
  Cloud Scheduler runs a job every 3 hours
- **Keys stay secret**
  Secret Manager and a dedicated service account
- **A new state is a config file**
  Same app, models and agents for any coast

WHAT'S NEXT

Next: from the control room to every coastal home.

SurgeSight for citizens

"Am I at risk?" in my language, the nearest safe shelter, and a lower-risk route to it.

Cascading failures

See which hospitals lose power or water when a substation or pumping station floods.

Straight to every phone

Send approved CAP alerts to NDMA's SACHET platform for cell broadcast.

Voice for the last mile

Village officers call in and ask Gemini Live in Tamil, Telugu or Odia.

Money before landfall

Parametric triggers release relief funds before the storm hits, not weeks after.

Learns from every storm

Satellite flood maps after each event retune the model. Then floods and heatwaves too.

WHY IT MATTERS

The right school, hospital and road protected first, before landfall.

District Collectors

A ranked list of what to protect first, with the reason for each.

Hospitals

Fuel, power and patient moves planned before the roads flood.

Power and water boards

Crews sent to the substations and pumps most at risk.

People on the coast

Warnings in their own language, from the officers they trust.

Michaung: 10.5 M people · 4,214 facilities · 7 departments · 6 languages

Screening, not forecasting · A person approves every alert

Scan to try SurgeSight

TEAM ECHO

Charan · Vishal

Live app

Demo video: youtu.be/BeFNW2ZWFQM

Code: gitlab.com/charanvelavan12/surgesight